



National University
Polyclinics

WEIGHT MANAGEMENT IN PRIMARY CARE

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I. INTRODUCTION

Weight is a clinical indicator and predictor of health. Obesity remains a major risk factor for non-communicable diseases such as cardiovascular diseases (CVD), stroke, diabetes mellitus (DM), musculoskeletal disorders and some cancers. Weight loss in obese patients reduces the risk of CVD and mortality due to its beneficial effects on blood pressure (BP), glucose intolerance and lipids. Effective weight management depends on a multi-disciplinary team providing holistic care. The key areas include diet, exercise (for obese / overweight patients), behavioral modification and medications (in some patients). **Importantly, care providers need to recognize obesity is a chronic condition with complex interactions that continues to have significant stigma and should be broached with empathy.**

However, progressive (involuntary) weight loss in adults should raise suspicion for a serious medical or psychiatric illness and is associated with increased mortality. Patients who are underweight may have medical complications.

II. OBESITY

Definition of Obesity (Based on Body Mass Index [BMI])

$$\text{BMI} = \frac{\text{Weight (kg)}}{\text{Height} \times \text{Height (m}^2\text{)}}$$

WHO classification of adults according to BMI¹

Category	BMI (kg / m ²)	Risk of Co-Morbidities
Underweight	< 18.5	*Low
Normal weight	18.5 – 24.9	Average
Overweight (Pre-obese)	25.0 – 29.9	Increased
Obese		
- Class 1	30.0 – 34.9	Moderate
- Class 2	35.0 – 39.9	Severe
- Class 3	≥ 40.0	Very severe

** Increased risk for other clinical problems, i.e., osteoporosis, subfertility, nutritional deficiencies*

Asian populations have a higher percentage of body fat, with corresponding higher prevalence of Type 2 DM and CVD risk factors compared to Caucasian populations of the same age, sex, and BMI.

The following are proposed BMI cut-off points for public health action in Asians (adapted from a WHO Report)²:

CVD Risk	Asian BMI Cut-off Points for Action (kg/m ²)	Current WHO BMI Cut-off Points (kg/m ²)
	< 18.5	< 18.5
Low	18.5 – 22.9	18.5 – 24.9
Moderate	23.0 – 27.4	25.0 – 29.9
High	27.5 – 32.4	30.0 – 34.9
Very high	32.5 – 37.4	35.0 – 39.9
	≥ 37.5	≥ 40.0

Revised Asian BMI cut-offs are used to identify at-risk individuals for screening of chronic diseases and positive lifestyle advocacy. All Asian patients aged 18 to 65 years identified to have a BMI of ≥ 23 kg/m² should be told that they are at higher risk of medical conditions such as DM, hypertension (HTN) etc. Screen them for CVD risk factors and manage these risk factors accordingly.

Waist Circumference

This is the most practical anthropometric measurement for assessing patient's abdominal fat content before and during weight loss treatment and should be used in conjunction with BMI to identify increased disease risk. Based on an Asian-Pacific consensus, cut-offs of 90 and 80 cm respectively for males and females may be more appropriate for Asians.

A high waist circumference increases the likelihood of cardio-metabolic risk factors such as Type 2 DM, lipid disorder or HTN³. To assess, measure the waist at the midpoint between the lower margin of the last palpable rib and the upper border of iliac crest, with the patient standing.

Definition of high-risk gender-specific waist circumference:

Guideline	Waist circumference (cm)	
	Men	Women
WHO ⁴	≥ 102	≥ 88
Asia-Pacific Consensus ⁵	≥ 90	≥ 80

Metabolic Syndrome

Defined as co-occurrence of metabolic risk factors for Type 2 DM and CVD, specifically abdominal obesity, hyperglycaemia, dyslipidaemia, and HTN. There are different definitions with The National Cholesterol Education Program (NCEP) Adult Treatment Panel III (ATP III) being the most widely used. The presence of any **three** out of the following would define metabolic syndrome⁶.

1. Abdominal obesity ≥ 90 cm in men and ≥ 80 cm in females (adopting Asian cut-offs)
2. Serum triglycerides ≥ 1.7 mmol/L or drug treatment for elevated triglycerides
3. Serum HDL cholesterol < 1.0 mmol/L in male or < 1.3 mmol/L in females, or drug treatment for low HDL-C
4. BP ≥ 130/85 mmHg or drug treatment for HTN
5. Fasting plasma glucose ≥ 5.6 mmol/L or drug treatment for elevated blood glucose

Metabolic syndrome is an important risk factor for subsequent development of Type 2 DM and/or CVD. Key clinical implication of its diagnosis is to identify a patient who needs aggressive lifestyle modification focused on weight reduction and increased physical activity. A patient who is not overweight or obese (BMI criteria) may still have metabolic syndrome thus putting him / her at a higher risk of developing Type 2 DM and CVD.

Screening and Approach to Obesity

1. Measure BMI routinely, and at least every 6-monthly
2. Consider measuring waist circumference.
3. Clinical assessment
 - Causes: focused dietary, lifestyle and sleep history, physical examination, investigations
 - Secondary causes are rare; they may include drug-induced, endocrinopathies (Cushing's, hypogonadism, hypothyroidism) and genetic disorders (from early childhood include Prader-Willi, melanocortin-4 receptor deficiency)
 - Psychological issues such as insomnia, depression and binge eating can contribute to obesity.
 - Complications
 - Risk factors
 - Readiness to lose weight.
4. Goal setting and therapy
 - Reduce body weight by 5 to 10% over 6 to 12 months.⁷
 - HSG-enrolled patients during health plan discussion: to initiate and document conversations on weight loss when appropriate and refer suitable patients to weight management programs.

Assessment for Obesity

Causes of Obesity

Primary

- Genetic (Polygenic)
- Environmental (often implicated)
 - Childhood weight history
 - Sedentary lifestyle
 - High intake of energy-dense foods
 - Lack of regular physical activity
 - Heavy marketing of energy-dense foods and proliferation of fast food
 - Sleep insufficiency
 - Socio-cultural, psychological triggers (e.g., significant life events, job change, smoking cessation), financial resources or lack of (impact on food choices)

Secondary (less common) *

- Single gene defects
- Chromosomal defects
e.g., Prader Willi, Laurence-Moon-Biedl
- Associated genes / polymorphisms.

Diseases*

- Cushing's syndrome
- Hypothyroidism
- Polycystic ovarian disease
- Hypogonadism
- Metabolic syndrome

Drugs (e.g.)

Antidepressants (e.g., mirtazapine, paroxetine, amitriptyline), Antipsychotics (e.g., olanzapine, quetiapine, risperidone), Beta-blockers (e.g., atenolol, metoprolol), Diabetic agents (insulin, sulfonylureas) lithium, diphenhydramine, steroids

Complications / Related conditions

- HTN
- Type 2 DM
- Dyslipidaemia
- Coronary Heart Disease / Stroke
- Osteoarthritis of the lower limbs
- Gallbladder disease, Non-alcoholic fatty liver, Non-alcoholic steatohepatitis
- Gout
- Cancers
 - Male – colorectal, prostate
 - Female – endometrial, gallbladder, cervical, ovarian and breast
- Obstructive sleep apnoea

Physical examination

- BMI
- Waist circumference
- BP
- Consider signs of insulin resistance (acanthosis nigricans), pathological causes (rounded facies, increased interscapular and supraclavicular adiposity, goitre)

Investigations (as appropriate):

- 1) Metabolic profiling
 - Lipid panel (TG, LDL-C, HDL-C)
 - Fasting plasma glucose
- 2) Aetiology / secondary causes*
 - Thyroid function test
 - Cushing's syndrome (only if clinically suspected)
 - Hypogonadism (only if clinically suspected)
- 3) Complications
 - ECG (if >50-years-old)
- 4) Others
 - e.g., Fitness for exercise program

Assess Readiness to Lose Weight

- Motivation for weight loss (important prerequisite)
- Stage of readiness: Pre-contemplation, Contemplation, Decision, Action, and Maintenance
- Previous attempt at weight loss
- Support from family and friends
- Understanding risks of being overweight / obese and the benefits of weight reduction
- Attitude towards physical activity
- Time availability
- Potential barriers

* Refer patients with secondary causes of obesity to the relevant specialty clinic

Management and Follow-Up

1. Patient education and lifestyle modification

- Concept: Multi-faceted and multi-disciplinary strategy should be utilised to achieve and maintain weight loss in obesity. Lifestyle modification should be the main treatment strategy for weight management.
- Inform overweight or obese patients of their condition.
- Risks of high BMI and related risk of CVD, Type 2 DM, and all-cause mortality.
- Benefits of weight loss.
- Patients with CVD risks, modest, sustained weight loss of 3-5% can produce clinically meaningful health benefits. Realistic weight loss should be safe and preferably not exceed 0.5-1kg a week.
- Aim for initial 5-10% weight loss over 6 months (if not attainable, a prevention of further weight gain may be recommended to improve mechanical symptoms and metabolic parameters). Long-term goal of more weight loss is encouraged (e.g., 10-20% of initial body weight).
- Use a combination of long-term dietary modification, increased physical activity and behavioural therapy.

2. Pharmacotherapy

- Consider when BMI is $\geq 30 \text{ kg/m}^2$, or when BMI is 27.5–29.9 kg/m^2 in Asians with comorbidities or complications of obesity such as HTN, Type 2 DM.
- Adjunct to management that includes exercise, dietary and behavioural modification.

3. Surgery

- Should be considered an option for those with severe obesity who have failed weight management programmes which include lifestyle change and pharmacotherapy.

4. Weight loss maintenance

- Provide regular follow-up. The patient is encouraged to continue a reduced calorie diet, maintain increased physical activity, and monitor their body weight regularly.

Goal Setting and Therapy

Setting Goals

- Realistic and achievable goals (e.g., reduction of body weight by 5 to 10% over 6 to 12 months).
- Maintain a lower weight over long term with prevention of further weight gain.
- Success measured by degree of weight loss and improvement in risk factors.

Management

- Healthy diet
- Physical activity
- Behavioural therapy
- Pharmacotherapy
- Surgery

1. Dietary Counselling

- Caloric reduction is the most important part of weight loss and maintenance. Aim for caloric deficit by 500 kcal below estimated daily energy requirement. This produces a recommended weight loss of 0.5 kg per week.
- A balanced, nutrient reduction diet seems most effective in longer-term weight maintenance (restricted in total fat, moderate to high in complex carbohydrates and moderate in protein). Long-term nutritional adjustments are important to sustain weight loss.
- Caution when recommending high-protein, low-CHO diet in patients with osteoporosis or kidney disease.
- Skipping of meals should be discouraged. Food portion at each meal should be adequate to prevent frequent snacking between meals. Sugar-sweetened beverage should be avoided.
- Maintain diet higher in vegetables, legumes, fruits, and whole grains, lower in fat, sugar, and refined carbs.
- Refer to a dietitian / trained healthcare professionals to educate the patient on meal planning and reading of food labels.

2. Physical Activity

- Regular physical activity is recommended to optimize health and protect against dementia, cardiovascular disease, cancer, and other non-communicable diseases.
- Use F.I.T.T to guide exercise – Frequency, intensity, time, and type.
- Perform moderately intense physical activity for:
 - At least 150 to 420 minutes per week for weight loss.
 - A longer duration of more than 300 minutes per week is needed for clinically significant weight loss and for weight maintenance after weight loss⁸.

- Higher intensity activity or exercise has better effects on weight loss compared to lower intensity activity.
- Use Karvonen's Formula to determine training heart rate zones: Training Heart Rate¹⁵ = ((Maximum Heart Rate – Resting Heart Rate) × Intensity%) + Resting Heart Rate. A minimum of 60% of intensity% is suggested for weight loss.
- Consider using 'talk test' (see table below) if there is difficulty calculating training heart rate zones.
- Compound exercises involving multiple muscle groups is encouraged.
- The exercise sessions can be done in a single continuous manner OR in multiple 10-15 min sessions daily to make up for at least 30 minutes.
- Examples of physical activities (can include both structured exercise sessions and incidental activities): household chores, walking for commute, brisk walking, marching on the spot, swimming, cycling or team sports.
- Reduce sedentary periods by getting up and moving for 2-5 minutes with every hour of sitting.
- For a patient who has been sedentary, he / she should start off by exercising 10 -20min, 2-3 times a week, for 2 weeks before gradually increasing either the duration or intensity of exercise.
- Contraindications to exercise
 - Absolute contraindications to exercise include recent / complicated myocardial infarction, unstable angina, uncontrolled dysrhythmias, uncontrolled HTN or metabolic disorders, moderate and severe valvular heart diseases e.g., severe aortic stenosis.
 - Relative contraindications are systemic infections which include viral fever.
 - Refer physiotherapist or a sports physician for exercise prescription if necessary.
 - Refer to a physiotherapist when patient's musculoskeletal pain, pre-existing medical condition or functional mobility reduces the patient's ability to be active.

	Aerobic Training	Resistive (Anaerobic) strength Training
Target and Time Spent	1. At least 150 min moderate intensity^α or 75 minutes vigorous intensity^β activity or 2. 300-420 minutes of moderate exercise or 150 minutes vigorous activity if tolerable for weight loss 3. Step count: ≥ 10,000 steps ^α Moderate intensity: can talk in sentences but cannot sing ^β Vigorous intensity: can speak only a few words in a breath	1. Exercise for core muscle group: sit up, planks, bridging, lunges, squats, push-ups. 2. Strengthening exercise for upper and lower limbs: lunges, squats, push- ups, bicep curls. 3. Balance between heavier weight and low repetitions to gain muscle mass or improving endurance by light weights with higher repetitions.

3. Behavioural Modification

Motivational interviewing (MI) is evidence-based and useful for patients ambivalent about weight loss as it helps move people through stages of change.

Ascertain a patient's stage of readiness to change and assist him / her in moving onto the next stage (Pre-contemplation > Contemplation > Preparation > Action > Maintenance). Consider his / her knowledge, attitudes, and beliefs while adopting a non-judgmental, individualised approach. Collaborate with the patient to modify behaviour through self-monitoring, stimulus control and social support. Consider referring to a psychologist for motivational interviewing to address ambivalence and effect lasting change.

- Self-esteem
 - Explore patient's issues and help patient to accept self.
- Stimulus control
 - Learn what cues encourage undesirable eating and change those cues (e.g., snacking while watching television)
 - Encourage behaviours that are incompatible with eating when there is an urge to overeat.
- Cognitive assessment
 - Identify negative thoughts and help patient reframe them.
- Stress management
 - Encourage other methods to deal with stress that do not involve eating.
- Behavioural goal setting
 - Understand that obesity is a chronic problem.
 - Set attainable goals, emphasize small steps, and reward self when goal is achieved.
 - Eating and activity patterns to change permanently to achieve modest and sustained changes in weight and risk factors.
- Self-monitoring
- Coping with relapses
 - Normalise setbacks and encourage to start afresh
- Realistic evaluation and rewards (not in the form of food)
- Social support
 - Family / healthcare providers / support group

Motivational Interviewing Techniques*		
Technique	Example	Rationale
Permission to discuss	"Would it be okay if we talk about your weight today?"	When given permission, he or she is more open to the conversation
Empathy	"Losing weight is tough"	Aids in building rapport
Scale motivation (0 [low] – 10 [high])	"On a scale, how motivated are you to try to lose weight?"	Assess motivation to change, if low, patient may not be ready for change
Scale confidence (0 [low] – 10 [high])	"On a scale, how confident are you to lose weight?"	Assess need for interventions to overcome obstacles
Inquire about scores on above scales	"Why did you choose 4 instead of 2"?	Further the conversation on thinking about change
Decision balance technique	"What are pros of losing weight?" "What are the cons of not losing weight?"	Helps patient and physician understand barriers and motivators for change
Listen for change talk and reinforce; allow patient to take ownership by generating ideas for change	Patient: "I think I could try to walk more." Provider: "That's a great idea that will help you move towards your goal."	Encourage and promote confidence in patients

*Adapted from AAFP Update on office-based strategies for the management of obesity, Sep 2016

4. Medications

Indications

BMI ≥ 30 kg/m² with or without obesity-related complications or risk factors **OR**

When BMI is 27.5 – 29.9 kg/m² among Asians with co-morbidities or complications such as HTN, Type 2 DM.

It is used in a programme that includes dietary modification, increased physical activity and behavioural therapy.

Use of drugs to achieve weight loss should be for medical indication only and not for cosmetic reasons. Drug therapy for weight management as an adjunct to lifestyle modification is Medisave claimable in the presence of CDMF diagnoses such as Type 2 DM, HTN, or hyperlipidemia.

Drug Name	Mode of Action	Dosing	Adverse Effects
Orlistat (Xenical)	Inhibits gastrointestinal lipases and blocks absorption of ingested fat	Consider as adjunct to diet and exercise in patients with BMI ≥ 30 kg/m ² or BMI ≥ 27 kg/m ² with CHD risk factor(s) 120 mg with each main meal (Up to three times daily) immediately before, during or up to one hour after each main meal. Omit dose if a meal is missed or contains no fat	Steatorrhea, bloating, flatulence with discharge and abdominal pain (risk of gastrointestinal events increases when taken with a meal very high in fat) Rare cases of liver injury reported. Contraindicated in pregnancy, breastfeeding, cholestasis, and chronic malabsorption syndrome. Consider prescribing a multivitamin containing fat-soluble vitamins, at bedtime or at least 2 hours before or after orlistat administration, in chronic use. Discontinue if weight loss is < 4% of baseline after 3 months.
Liraglutide (Not available in NUP) Semaglutide (Not available in NUP)	Glucagon-like peptide-1 (GLP-1) analogue with anorexigenic effect	Liraglutide: SC 0.6 mg once daily (Up to 3 mg once daily) Semaglutide: SC 0.25 mg once weekly (Up to 2.4 mg once weekly) As an adjunct to a reduced-calorie diet and increased physical activity for weight management in adults with an initial BMI of ≥ 30 kg/m ² or ≥ 27 kg/m ² in the presence of at least one weight-related comorbidity e.g., dysglycaemia (prediabetes or type 2 diabetes mellitus), hypertension, dyslipidaemia, obstructive sleep apnoea or cardiovascular disease.	Nausea, vomiting, diarrhoea, constipation, and abdominal pain. Rare associations with kidney injury, gallbladder disease and pancreatitis reported. Consider in patients with T2DM and weight loss is one of the goals. Contraindicated in pregnancy, personal or family history of medullary thyroid carcinoma. Treatment with Liraglutide should be discontinued after 12 weeks on the 3.0 mg/day dose if weight loss is < 5% of initial body weight.

Drug Name	Mode of Action	Dosing	Adverse Effects
Phentermine (Not available in NUP)	Enhances satiety and suppresses appetite through noradrenergic pathway	15 mg once daily at breakfast (Up to 30 mg once daily) Avoid evening dose (may cause insomnia) For short term use as an adjunct to the treatment of moderate to severe obesity	HTN, palpitations, tachycardia, tremors, headache, insomnia, restlessness, gastrointestinal distress and change in libido. Contraindicated in pregnancy, breastfeeding, history of cardiovascular disease, hyperthyroidism, glaucoma, agitated states, history of drug abuse. Has abuse potential, prolonged use may lead to dependency. Tolerance to the anorectic effect usually develops within a few weeks; discontinue use when tolerance develops, do not exceed recommended dosage to overcome tolerance.
Naltrexone-bupropion (Contrave) (Not available in NUP)	Act on areas of the brain involved in the regulation of food intake: the hypothalamus (appetite regulatory centre) and the mesolimbic dopamine circuit (reward system)	Initiate at naltrexone 8 mg/bupropion 90 mg OM. Increase as tolerated in weekly intervals up to 2 tablets BD. Do not take with high fat meal (significant increase in bupropion and naltrexone systemic exposure) As an adjunct to a reduced-calorie diet and increased physical activity for chronic weight management in adults with an initial BMI of $\geq 30 \text{ kg/m}^2$ or $\geq 27 \text{ kg/m}^2$ in the presence of at least one weight-related comorbid condition	Headache, nausea, vomiting, diarrhoea, constipation, hypertension, dizziness, insomnia. Contraindicated in uncontrolled hypertension, seizure disorder, chronic opioid, opiate agonist or partial agonist use, severe renal or hepatic impairment. Monitor for worsening or emergence of suicidal thoughts and behaviours. Re-evaluate treatment if lost $< 5\%$ of baseline body weight after 12 weeks. Drug interactions with Ticlopidine and Clopidogrel, maximum Contrave dose should not exceed 1 tablet twice daily when co-administered.

5. Surgery

Indications

Clinically severe obesity

BMI $\geq 40 \text{ kg/m}^2$

BMI $\geq 35 \text{ kg/m}^2$ with obesity related conditions*

Commensurate BMI thresholds for action among Asians $\geq 37.5 \text{ kg/m}^2$ and $\geq 32.5 \text{ kg/m}^2$ respectively.

Mechanism of weight loss

Bariatric surgery affects weight loss through two mechanisms: (1) restriction and (2) malabsorption. Some procedures have both a restrictive and malabsorptive component. They may also contribute to neurohormonal effects on the regulation of energy balance.

Types⁹

- 1) Restrictive e.g., sleeve gastrectomy, laparoscopic adjustable gastric band
- 2) Malabsorptive e.g., biliopancreatic diversion, biliopancreatic diversion with duodenal switch
- 3) Combination of restrictive and malabsorptive e.g., Roux-en-Y gastric bypass

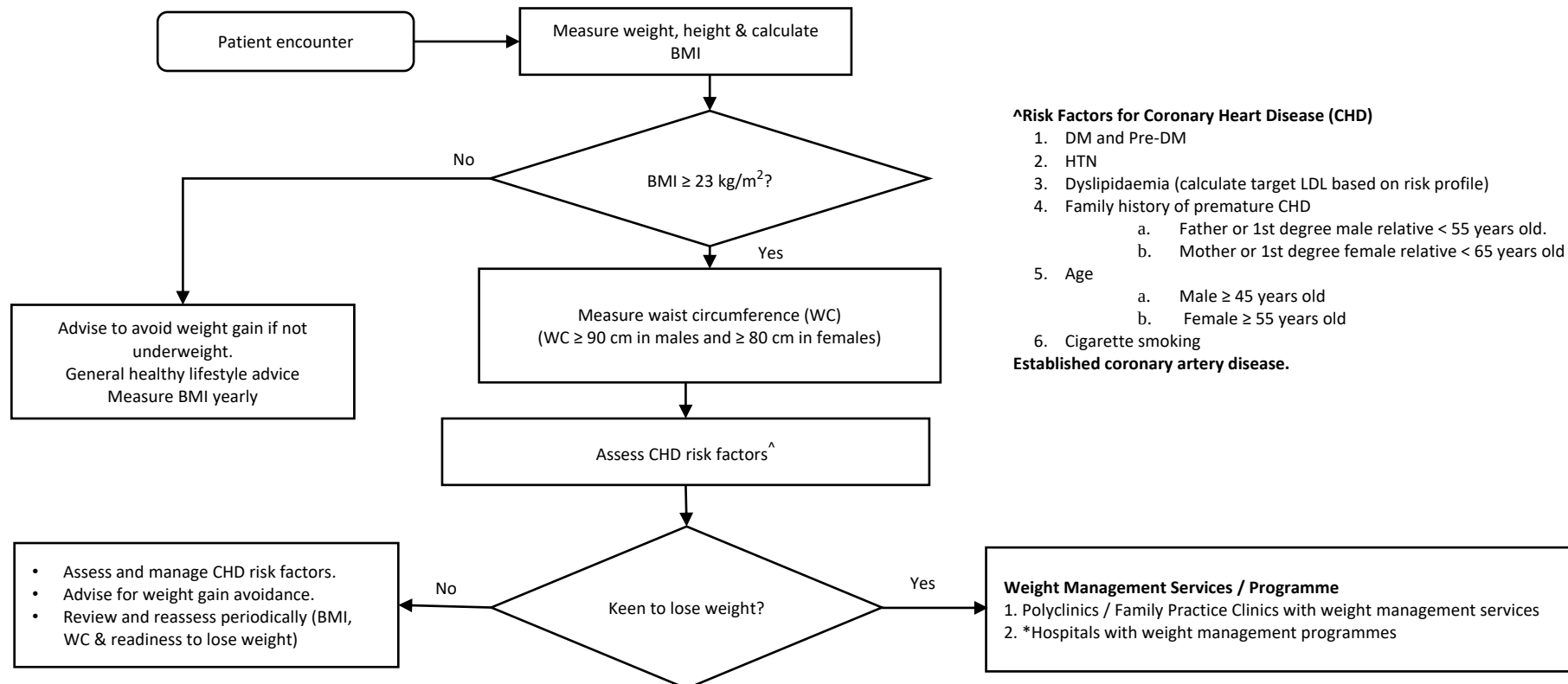
Contraindications

- 1) Elevated risk for surgery
- 2) Major psychiatric illness
- 3) Significant comorbidities
 - Endocrine causes for obesity (Cushing's syndrome, hypothyroidism)
 - Gastro-intestinal and hepatobiliary diseases (IBD, chronic pancreatitis, liver cirrhosis, portal HTN, malignancy)
- 4) Regular NSAID or corticosteroid use
- 5) Alcohol or substance abuse
- 6) Pregnancy
- 7) Children

*Obesity related conditions include Type 2 DM, HTN, hyperlipidaemia, fatty liver, polycystic ovarian syndrome, obstructive sleep apnoea and metabolic syndrome

Weight Management Programme for Adults in a Primary Healthcare Setting (Singapore)

(Age 18 to 65 years old)



*Suggested criteria for referral to a hospital-based Weight Management Clinic. (Referral criteria for different hospitals may vary)

- BMI ≥ 27.5 kg/m² with CHD risk factors. (This can be managed at the primary healthcare level first)
- BMI ≥ 37.5 kg/m² with no risk factors or BMI ≥ 32.5 kg/m² with CHD risk factors for consideration of surgical therapy
- Consider in failure to reach target weight loss (5 to 10% over 6-12 months) or weight gain after 1 year at the primary healthcare level.

Exclusion criteria (for NTFGH weight management clinic)

- eGFR < 30 ml/min/1.73m²
- History of substance abuse
- Eating disorder
- Difficult to control mood and psychotic disorder.
- Significant cardiac disease, MI within 6 months, NYHA grade 3 or more, angina /arrhythmia
- Recent stroke within the last 6 months
- Pregnancy
- Breast feeding
- Significant mobility issue, e.g., presence of limb weakness, moderate to severe deformity, contractures OR wheelchair bound most of the time.

III. UNINTENTIONAL WEIGHT LOSS

Approach to Underweight Patients¹⁰

Patients who are underweight (BMI < 18.5 kg/m²) or have unintentional weight loss (documented weight loss of > 5% of usual body weight over 6-12 months excluding weight loss as an expected consequence of treatment or a result of a known illness) should be evaluated clinically and managed accordingly. These patients are at higher risks of medical problems such as nutritional deficiency, osteoporosis, and sub-fertility. Look for these complications and manage accordingly.

Voluntary weight loss in overweight or obese individuals is benign in most instances.

For patients with involuntary weight loss, it can indicate an underlying serious medical or psychiatric illness.

Major Causes of Weight Loss

Unintentional weight loss	Voluntary weight loss
Medical disorders a) Malignancy (e.g., gastrointestinal, lung, lymphoma, renal and prostate etc.) b) Gastrointestinal diseases (e.g., malabsorption, peptic ulcer disease, inflammatory bowel disease) c) Endocrine diseases (e.g., hyperthyroidism, Type 2 DM) d) Infectious diseases (e.g., HIV, viral hepatitis, tuberculosis, chronic fungal or bacterial disease, chronic helminth infection) e) Advanced chronic diseases (heart, lung, or kidney disease) f) Neurological diseases (stroke, dementia, Parkinson disease, amyotrophic lateral sclerosis) Psychiatric disorders Affective disorders (e.g., depression, bipolar disorder) Drugs a) Alcohol b) Opiates, amphetamines & cocaine c) Adverse effects of drugs (e.g., SSRIs, levodopa, metformin) d) Herbal & other nonprescription drugs (e.g., 5-hydroxytryptophan, caffeine)	Drugs a) Drugs for treating obesity (e.g. rimonabant, orlistat, phentermine, liraglutide) b) Drugs occasionally abused for weight loss effects (amphetamines & derivatives, thyroid hormone) c) Herbal & other nonprescription drugs (5-hydroxytryptophan, caffeine, chitosan) d) Bupropion, SGLT2 inhibitors, topiramate e) Tobacco / Smoking Dieting a) Anorexia nervosa & bulimia nervosa b) Chronic vigorous exercise, usually combined with dieting (e.g., distance runners, ballet dancers, gymnastics)

Causes of Weight Loss in Older Adults

M	Medications (e.g., digoxin, theophylline, SSRIs, antibiotics) + Malignancy
E	Emotional (e.g., depression)
A	Alcohol-related disease, (older) adult abuse, Anorexia nervosa
L	Late-life paranoia
S	Swallowing problems, systemic issues (chronic cardiopulmonary, renal, endocrine, autoimmune pathologies)
O	Oral factors (poor fitting dentures, caries)
N	Nosocomial infections (e.g., tuberculosis, pneumonia), No money
W	Wandering and other dementia-related behaviors
H	Hyperthyroidism, hypercalcemia, hyperglycemia, hypoadrenalism, HIV
E	Enteric problems (e.g., esophageal stricture, gluten enteropathy)
E	Eating problems
L	Low salt, low cholesterol diets
S	Stones (cholecystitis), Social factors (shopping and meal preparation problems)

Evaluation (Weight Loss)

Baseline evaluation for unintentional weight loss includes:

- History and physical examination focusing on cause of weight loss.
- Targeted geriatric assessment assessing cognitive impairment, depression and functional status.
- Medication and supplement use.
- Nutritional impairment screening (e.g., Mini nutritional assessment: <https://www.mna-elderly.com>)
- Body weight without shoes.
- Physical examination of oral cavity, dentition and focused examination for heart, lung, gastrointestinal, or neurologic abnormalities depending on history.

Symptoms	Possible diagnosis	Suggested tests
Dysphagia, oral/gum problems	Poorly fitting dentures, dental caries, periodontal disease, esophageal strictures	Visual assessment (Swallowing study – only as indicated)
Dyspnea, exertional fatigue	Heart failure, Lung infection, COPD	FBC, possible CXR, ECG, metabolic screen
Fever	Infection, autoimmune disorders, malignancy, DM, thyroid disorder	FBC, ESR, HbA1c, thyroid function
Indigestion, change in bowel habits	Gastrointestinal malignancies, peptic ulcer disease, GERD, chronic cholecystitis	FBC, liver function, US HBS, colonoscopy

Shared decision-making and goals of care should guide the above evaluation. Appropriate follow-up of patients with unexplained and unintentional weight loss is needed and further referral for workup is recommended if initial tests are negative with no mental health issues.

Treatment and Management

- Focus on underlying causes and multidisciplinary team (dentist, dietitian, speech, occupational, physiotherapist and/or social worker) can be involved (as appropriate).
- Address dietary changes, environmental modifications, and nutritional supplements.
- Avoid appetite stimulants and high-calorie supplements due to lack of long-term survival benefits.

Approach to Weight Loss Management

Documented weight loss 5% or more of usual body weight over 6-12 months



Targeted history:

Systemic review, past medical and psychiatric (eating disorder) history, dietary, social, functional and exercise history.
Pattern of weight loss and intentionality of weight loss
Associated symptoms (e.g., malignancy, malabsorption)
Medication and substance (such as amphetamines, tobacco) use

Physical examination:

Targeted examination starting with general appearance and mental state review. Look for signs suggestive of malignancy, endocrine, gastrointestinal, cardiopulmonary, infectious, nutritional, neurologic & psychiatric illnesses.



Investigations (order *only according to* suspected diagnosis, based on history and physical examination)

1. FBC
2. Thyroid function test (free T4, TSH)
3. Plasma glucose / HbA1C
4. Erythrocyte sedimentation rate
5. Creatinine
6. Liver function test
7. Chest X-ray (looking for tuberculosis, chronic suppurative / obstructive lung disease)
8. HIV, Hepatitis C (as clinically appropriate)
9. Age-appropriate cancer screening (FIT or colonoscopy, mammography, cervical screen)



Manage accordingly based on findings.
May need referral to specialist outpatient clinics for further investigations.

IV. ROLE OF HEALTH TEAM MEMBERS

1. Family Physician

- Able to provide all aspects of primary medical care from screening for obesity or unintentional weight loss, diagnosis, and management of weight management, including health promotion and prevention / treatment of complications.
- Collaborates with other health care providers in the health team to provide holistic and patient-centered care.

2. Care Manager

- Empower patients through health promotion and disease prevention.
- Provide tailored education on weight management, disease process and lifestyle changes.
- Collaborate with other health care providers to develop care plans to provide patient-centered care.

3. Advanced Practice Nurse

- Collaborates with physicians to develop plan of care for weight management to prevent / treat complications.
- Collaborates with other health care providers to provide holistic care for patients.

4. Pharmacist

- Smoking cessation clinic
- Detect, prevent, and address drug-drug / drug-disease interactions.
- Perform medication reconciliation.

5. Dietitian

- Dietary management for patients with BMI >23 – goal setting and meal planning.
- Patient education on weight maintenance after weight loss.
- Supporting individuals in making sustainable changes to their dietary habits.
- Dietary management for patients with unintentional weight loss requiring medical nutrition therapy to prevent and/or manage malnutrition.
- Dietary management for children who are underweight or overweight based on BMI percentile <10% and >90% respectively.

6. Medical Social Worker

- Providing holistic support to patients struggling with obesity and weight loss by addressing their physical, emotional, and social needs.
- Counselling and education on healthy lifestyle choices, nutrition, and exercise

- Helping patients to access community resources such as Active Ageing Centre, support groups and programs that promote healthy living and weight management.

7. Psychologist

- Psychological and behavioral interventions to address barriers to health behavior change and maintenance.
- Psychological assessment and intervention to manage co-occurring psychological problems (eating disorders, depression, anxiety disorders etc.)

8. Physiotherapists

- Exercise prescription involving aerobic and resistance exercises.
- Education on calculating training heart rate zones.
- Education on use of external devices to track physical activity and the caloric expenditure of different types of physical activities.

9. Podiatrist

- Biomechanical assessment for in-shoe orthotic devices if any foot pain.
- Patient education on appropriate footwear options

V. SPECIAL SITUATIONS

Fertility and Pregnancy

- Females of childbearing age, especially overweight and obese women, should receive information and advice from healthcare providers about the benefits of weight loss before pregnancy and the risks of being overweight or obese. An ideal preconception weight will optimise pregnancy outcome for both mother and baby.
- Prior to attempting to conceive, females should stop taking medications for weight loss.
- Nutritional advice should be provided to pregnant females (“Pregnancy and Diet, HPB, Singapore”); include foods rich in calcium and iron.
- Post-bariatric surgery: Female patients should avoid getting pregnant until weight loss has stabilised (in general 12 to 24 months post procedure). Contraceptive advice should be given (oral contraception may be less effective due to malabsorption). Nutritional supplements over and above usual requirements in a normal pregnancy thereafter are recommended.

Children and Adolescents

- During transition to young adulthood, there may be discrepancy in weight status due to differences in the two classification systems. For older adolescents (e.g., >16 years old), obesity may be defined as a BMI-for-age equal to or greater than 97th percentile, or BMI equal to or greater than 30 kg/m², whichever criterion is met.
- Vast majority (> 95%) will have common or primary obesity. Investigations for comorbidities should not be regarded as routine but may be ordered as indicated clinically. Screening may include glucose levels, lipid profile, liver function test if clinically indicated.
- Obese children and adolescents should enrol in multi-component lifestyle interventions, which consist of nutrition, physical activity, and behavioural modifications, which can result in a modest, but significant reduction in obesity and CVD risk, compared to standard care, self-help, or no treatment.
- Parental involvement can enhance the effectiveness of lifestyle interventions, particularly in pre-adolescent children, by changing and adapting parenting styles, parenting skills and child management strategies during the intervention, such as role modelling and active parental participation.
- Pharmacotherapy for weight loss should **only** be considered as an adjunct to lifestyle interventions in obese adolescents with severe comorbidities or when lifestyle interventions have failed. It should be administered by experienced clinicians. Decision should be weighed against the potential for adverse effects, the lack of evidence for persistent weight loss after active treatment and the long-term safety in adolescents.

VI. REFERRALS AND COMMUNITY RESOURCES

- Children and adolescents: Refer to tertiary general paediatric, paediatric endocrine or medical genetic clinics are guided by the degree of obesity or severely overweight individuals with comorbidities, or where a pathological cause of obesity is suspected.
- Adults: Refer to Endocrine (weight management clinic, NTFGH) or bariatric surgery for medical and surgical weight management interventions respectively for patients who have severe obesity or obesity-related comorbidities

VII. ABBREVIATIONS

BP	Blood Pressure
BMI	Body Mass Index
COPD	Chronic Obstructive Pulmonary Disease
CVD	Cardiovascular diseases
CXR	Chest X-ray
DM	Diabetes Mellitus
ECG	Electrocardiogram
FBC	Full blood count
HTN	Hypertension

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